

Python Programming Lab

Lab Task: Data Types and Operators

Course	Python Programming
Topic	Data Types (Integers, Strings, Booleans) & Operators
Duration	2 Hours
Name / Roll No.	_____

Learning Objectives

- Understand and use the Integer, String, and Boolean data types in Python.
- Apply Arithmetic, Comparison, Assignment, Logical, Bitwise, Membership, and Identity operators.
- Write short Python programs and predict/verify their output.

Instructions

- Write and run each program in a Python IDE (IDLE, VS Code, Jupyter, or an online compiler).
- For every task, note down the code, the output, and your observations.
- Attempt all tasks in the given order. Do not skip data type conversion steps.

PART A: DATA TYPES

A1. Integers (int)

Integers represent whole numbers (positive, negative, or zero) without a decimal point.

Task A1.1: Write a program that:

- 1 Stores your age, the current year, and your birth year in three integer variables.
- 2 Prints the type of each variable using `type()`.
- 3 Calculates and prints your age in the year 2050.

```
age = 21
current_year = 2026
birth_year = current_year - age
print(type(age))
print("Age in 2050:", 2050 - birth_year)
```

Task A1.2: Predict the output, then verify by running the code:

```
x = 17
y = 5
print(x // y)    # integer division
print(x % y)     # modulus
print(x ** 2)    # exponent
```

A2. Strings (str)

Strings are sequences of characters enclosed in single, double, or triple quotes.

Task A2.1: Write a program that:

- 1 Stores your first name and last name in two separate string variables.
- 2 Concatenates them into a full name with a space in between.
- 3 Prints the full name in UPPERCASE, lowercase, and Title Case.
- 4 Prints the length of the full name.
- 5 Prints the first character and the last character of the full name.

```
first = "Ada"
last = "Lovelace"
full_name = first + " " + last
print(full_name.upper())
print(full_name.lower())
print(full_name.title())
print(len(full_name))
print(full_name[0], full_name[-1])
```

Task A2.2: Use string slicing to extract and print only your first name from the full name string (without using the original variable).

A3. Booleans (bool)

Booleans represent one of two values: True or False, often produced by comparisons or logical operations.

Task A3.1: Write a program that:

- 1 Declares two boolean variables, `is_raining` and `has_umbrella`.
- 2 Prints their type using `type()`.
- 3 Prints the result of `is_raining` and `has_umbrella`, `is_raining` or `has_umbrella`, and `not is_raining`.

```
is_raining = True
has_umbrella = False
print(type(is_raining))
print(is_raining and has_umbrella)
print(is_raining or has_umbrella)
print(not is_raining)
```

Note: In Python, bool is technically a subtype of int (`True == 1`, `False == 0`). Verify this by printing `True + True` and `False * 5`.

PART B: OPERATORS

B1. Arithmetic Operators

Operators: +, -, *, /, // (floor division), % (modulus), ** (exponent)

Task B1.1: Take two integers a = 23 and b = 6. Write a program to compute and print the result of all seven arithmetic operators listed above on a and b.

B2. Comparison / Relational Operators

Operators: ==, !=, >, <, >=, <=

Task B2.1: Take two integers m and n entered by the user (use input() and int()). Print the result of comparing them with all six comparison operators.

```
m = int(input("Enter m: "))
n = int(input("Enter n: "))
print(m == n)
print(m != n)
print(m > n)
print(m < n)
print(m >= n)
print(m <= n)
```

B3. Assignment Operators

Operators: =, +=, -=, *=, /=, //=, %=, **=

Task B3.1: Start with a variable score = 50. Apply each assignment operator one after another (in the order listed above), printing the value of score after every step.

B4. Logical Operators

Operators: and, or, not

Task B4.1: A student is eligible for a scholarship if their percentage is above 75 AND attendance is above 90%. Write a program that takes percentage and attendance as input and prints whether the student is eligible, using logical operators.

```
percentage = float(input("Enter percentage: "))
attendance = float(input("Enter attendance %: "))
eligible = percentage > 75 and attendance > 90
print("Eligible for scholarship:", eligible)
```

B5. Bitwise Operators

Operators: & (AND), | (OR), ^ (XOR), ~ (NOT), << (left shift), >> (right shift)

Task B5.1: Take two integers p = 12 and q = 10. Print the result of p & q, p | q, p ^ q, ~p, p << 2, and p >> 2. Convert p and q to their binary form using bin() and verify the bitwise results manually.

```
p = 12
q = 10
print(bin(p), bin(q))
print(p & q)
print(p | q)
print(p ^ q)
print(~p)
print(p << 2)
print(p >> 2)
```

B6. Membership Operators

Operators: in , not in

Task B6.1: Create a list of five fruits. Ask the user to enter a fruit name and check whether it is in the list or not, using the in and not in operators.

```
fruits = ["apple", "banana", "mango", "grape", "kiwi"]
item = input("Enter a fruit: ")
print(item, "is in the list:", item in fruits)
print(item, "is not in the list:", item not in fruits)
```

B7. Identity Operators

Operators: is , is not

Task B7.1: Create two lists with the same content, list1 and list2, and a third variable list3 = list1. Use the is and is not operators (along with id()) to check and explain which variables refer to the same object in memory.

```
list1 = [1, 2, 3]
list2 = [1, 2, 3]
list3 = list1
print(list1 == list2) # same content
print(list1 is list2) # same object?
print(list1 is list3) # same object?
print(id(list1), id(list2), id(list3))
```

Note: == checks for equal values, while is checks for the same object identity in memory.

Challenge Task (Optional)

Write a single Python program for a simple login validator: it stores a predefined username (string) and password (string), takes user input for both, uses comparison and logical operators to check the credentials, and uses a boolean variable login_success to store and print the final result.

Submission Checklist

Task	Completed (Y/N)	Output Verified (Y/N)
A1 - A3: Data Types		
B1 - B7: Operators		
Challenge Task		